Technical Portfolio

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1 Overview

I am a physics graduate (MPhys) from Durham University. I have completed several projects throughout my time there which I will be detailing in this portfolio. Code and writeups for my final two years may be viewed on my Github.

Skills Overview

Masters Project

- Python coding skills - learning and applying complex physics Python packages - pairinteraction and ARC
- Presentation skills - weekly results updates presentations and compiling results in a thesis
- Data analysis and presentation - visualising data in different ways and analysis - matplotlib

Summer Studentship

- Practical optical alignment and fine adjustment, psf optimisation and aberration correction
- Repeatibility testing and troubleshooting

Atmospheric Muon Lab

- Modular electronics and applications for data collection
- Design innovation and testing
- Prolonged data collection

Quantum computing Project

- Python programming - NumPy, SciPy, Matplotlib
- Building of quantum gates from basic python packages

Fourier Optics Lab

- Lens alignment and optics manipulation
- Fundamental optics principles

2 Master's Project - Characterising Dipolar Interactions between Rubidium and Caesium Rydberg Atoms (2025-2026)

2.1 Project Overview

- Finding electronic states for rubidium and caesium to inform a dual-species neutral atom quantum computing framework. Anand and et al. [2024]
- Specifically looking into Förster resonances in pairs of atoms.
- Using the context of a two-dimensional dual-species array to inform interaction strengths and minimise information loss to surrounding atoms.
- Understanding how these interactions may be used to create quantum gates to process information.

2.2 My Approach

- Using Python packages such as pairinteraction Mögerle et al. [2026] and ARC Šibalić et al. [2017] to calculate pair state energies and dipole matrix elements.
- Calculating energy defects between allowed pair states to search for resonant Förster interactions.
- Calculating dipole matrix elements for each pair state to use as a metric for interaction strength.
- Investigating the effect of using dc fields to tune a resonant interaction.
- Modelling how interaction for different separations and orientations.
- Investigating asymmetries in inter- and inter- species interactions to minimise the leakage of information through the array.

2.3 Result

- d-state pairs provide greater asymmetry than s-states, providing less information loss.
- A tradeoff lattice spacing needs to be reached to balance asymmetry and interaction strength.
- The pair state I propose is $|Rb : 53d_{3/2}, Cs : 60d_{5/2}\rangle$ for strong interactions and high asymmetry.
- I also report that high fields may be used to simplify the angular eigenstate landscape
- I find that both electric and magnetic fields may be used to separate magnetic sublevels then tune a certain pair to resonance, shown graphically in Fig. 1.

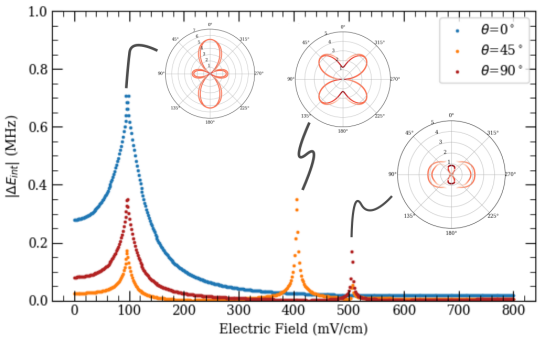


Figure 1: Electric tuning of a resonant interaction, insets show interaction shapes for different angles might look like for a given peak.

Master's Project References

S. Anand and et al. A dual-species Rydberg array. *Nat. Phys.*, 20(1744-1750), 2024. doi:10.1038/s41567-024-02638-2.

Johannes Mögerle et al. Accurate Modeling of Rydberg Atoms and Their Interactions: Theory and Implementation in PairInteraction, 2026. URL https://arxiv.org/abs/2605.14993.

N. Šibalić et al. ARC: An open-source library for calculating properties of alkali Rydberg atoms. *Comput. Phys. Commun.*, 220, 2017. doi:10.48550/arXiv.1612.05529.

3 Summer Studentship - Characterising a five-lens objective for Imaging of a Cs and Yb atom cloud (2025)

3.1 Project Overview

- Build on previous work for a home-built lens objective.
- I was initially adjusting spacings between each of the three casings around a Zemax-calculated value.
- Use python software, PyTOF Hoffman [accessed 2025], for calculating point spread functions and Zernike polynomials to make adjustments.

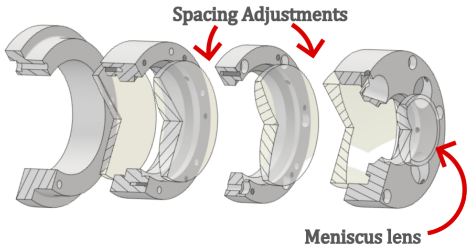


Figure 2: Lens design for this five-lens objective, consisting of four Thorlabs stock lenses and one custom meniscus lens. Drawing provided by Dr. Franzen / CsYb lab Durham University.

3.2 My Contribution

- Making adjustments to the lens spacings.
- Imaging a test slide with an array of 200nm pinholes using a low power green laser to characterise the point spread function of the objective.
- Aligning of the lens for testing different spacing and removing coma.
- Applying existing Python software to get a useable output for lens performance.
- Creating a method for reproducing the setup for when the lens will be installed in the experiment.

3.3 Result

- Achieved the desired resolution by setting the lenses in optimal separations and alignment of lens components.
- Created a system for moving the lenses independently in a controlled way to allow for reproducibility
- The system was designed to be set up in a repeatable way to remove coma.
- Made fine adjustments to spacings that would minimise primary spherical aberration.
- Aberrations calculated using Pytof software. Hoffman [accessed 2025] We see from Fig. 3 that the spherical aberrations were brought down to much less than 0.1 of a wavelength.

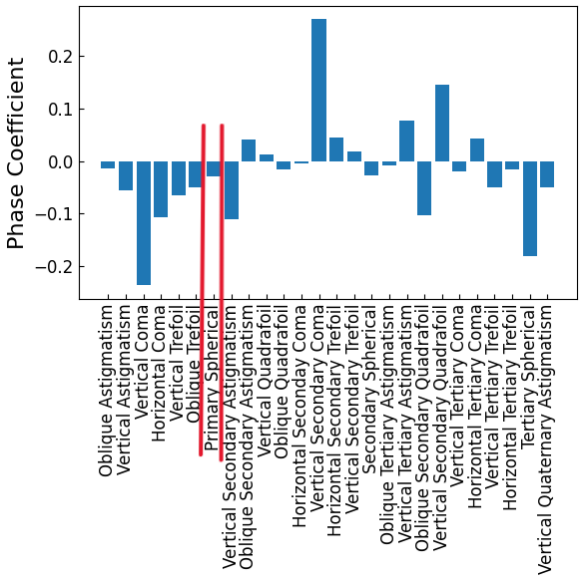


Figure 3: Zernike decomposition of the best image achieved, primary spherical is highlighted as the focus of this project, reduced by adjusting lens spacings.

Summer Studentship References

David P. Hoffman. pyotf: Python functions for simulating optical transfer functions. accessed 2025. URL https://github.com/david-hoffman/pyotf.

4 Level 3 Lab Project - Investigating the Lifetime of atmospheric Muons (2025)

4.1 Project Overview

- We were given a 'muon cupboard' apparatus for detecting muons comprisong of three layers of plastic scintillator material.
- This was used to detect muons formed in the atmosphere created by cosmic rays.
- We used three scintillators connected to photomultiplier tubes to output a pulse when a muon interacts with the scintillator.
- Several electrical (NIM) modules were at our disposal to process these signals into useable data, including discriminators and Time-to-Amplitude converters (TACs).

4.2 My Approach

- Characterise the scintillators to identify sources of noise.
- Engineer an electronic system for detecting multiple detector signals in coincidence - a sign that a muon passed through the apparatus.
- Set up timing apparatus to detect a muon interaction and a decay in the apparatus. Measurement process is shown in Fig. 4.
- Use prolonged data collection to give an estimate for the muon lifetime.
- Improve on the apparatus by installing a new, thicker piece of scintillator material to improve the chance of a decay in the apparatus.

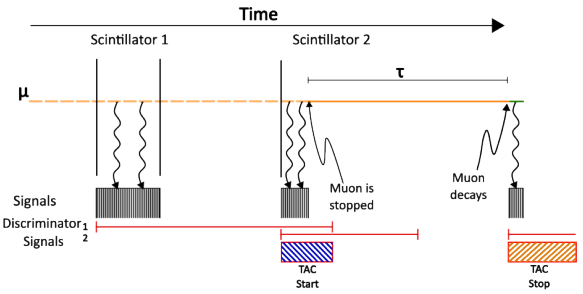


Figure 4: The process occuring when measuring a muon lifetime, it passes through the first scintillator, stops in the second and decays and we could use coincidence units to isolate these three signals.

4.3 Results

- Three different approaches to calculating muon lifetime were used to compare results.
- The simple single coincidence gave a value of $1.82 \pm 0.01\mu$s, the anti-coincidence gave of $2.00 \pm 0.01\mu$s the pulse generators returned a value of $1.59 \pm 0.02\mu$s.
- Comparing to the literature value of 2.197μs, Takashi et al. [2026] we get fairly close accounting for the time it takes to reach the detector.

Muon Experiment References

F. Takashi et al. The Review of Particle Physics (2026). *Int. J. Mod Phys. A*, 41(2630011), 2026.

5 Level 3 Computing Project - Modelling Basic Quantum Gates in Python (2024-2025)

5.1 Project Overview

- Build an understanding of basic quantum operations
- Develop code to simulate an addressing laser on a quantum computing framework, resulting in state transfer.
- Build basic quantum gate operations and investigate parameters that may affect these gate fidelities.
- Use physics Python packages like QuTip for simulations.

5.2 My Approach

- Initially I was researching basics of state transfer using lasers and their application to quantum computing.
- Coding the matrix transformations that describe the affect the addressing laser has on the atom state.
- Tailored different matrix transformations to execute the basic Hadamard gate.
- Investigated the specific series of pulses to achieve a CZ gate
- Considered real-world laser limitations such as pulse shapes.
- Considered effect of decoherence of atomic state and its affect on gate fidelity.

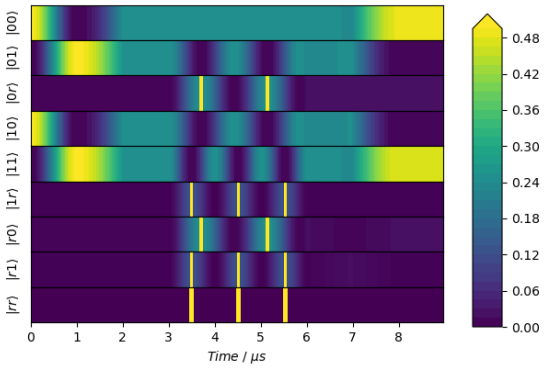


Figure 5: Broken down pair state populations for a CNOT pulse sequence. The result of which is the population of $|10\rangle$ moving to $|11\rangle$.

5.3 Results

- Fig. 5 shows a broken down population diagram for a CNOT gate which swaps the state of the second atom when the first atom is in state $|1\rangle$.
- I reported gate fildelities of $1 - 1.2 \times 10^{-4}$ for square pulses, $1 - 4.1 \times 10^{-8}$ for Gaussian and $1 - 10^{-15}$ for a more specially formulated pulse typically used for rapid adiabatic passage.
- Found that using the third shape of pulse returned the best fidelities.
- Different intensities and durations of pulses returned different fidelities.
- Plotted the effect of increasing spontaneous emission rates on final gate fidelities.

Interestingly this links to my final year project, where I focussed on the interaction while in the Rydberg state $|rr\rangle$. We can see the fast decay of the population in this state - in the centre of Fig. 5 - which allows for these quantum gates. Bosch et al. [2019]

Computing Project References

L. Yagüe Bosch et al. Shortcut-to-Adiabatic Controlled-Phase Gate in Rydberg Atoms. *Phys. Rev. Applied*, 11(034030), 2019.

6 Level 2 Lab Project - Investigating Fourier Optics via the use of a 4-f System (2023)

6.1 Project Overview

This project was undertaken as my year 2 research investigation. The project apparatus and task were laid out as follows:

- We were given a low power laser of several different wavelengths and access to optical components to build a 4-f (each length being one lens focal length) system.
- We had access to a variety of optical filters which we could apply in different positions in the system to observe the output.
- The goal of this project was to understand and verify Fourier transforms of objects when passed through lenses. Adams and Hughes [2018]
- The optics were setup as shown in Fig. 6.

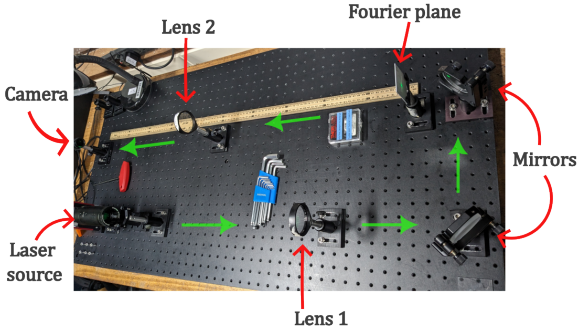


Figure 6: 4-f apparatus for investigating optical Fourier transforms.

6.2 My Contribution

My lab partner and I collaborated on the following steps:

- We chose an asterisk shape as the input function for most of our experiments
- Placing a single shape in the input plane and observing the Fourier transform in the Fourier plane.
- Placing a diffraction grating in the Fourier plane with an asterisk as the input function.
- Measuring the intensity of a point on the camera while displacing the diffraction grating.
- Fitting to a sine wave in order to calculate the slit spacing of the grating.
- Compared our fitted slit spacing measurement to the direct slit spacing when the grating is used as the input function.

6.3 The Result

- We reported slit spacing values much in agreement for three different gratings used.
- We used the lens of focal length 20cm as it returned better reduced χ^2 fitting values.
- We found a good relationship between image copy separation and wavelength of laser using our calculated slit spacing values and relationship in Eq. (1).

The relationship we were testing was

$$x = \frac{\lambda f}{d}, \tag{1}$$

where x is the position on the output plane, λ is the wavelength, f the focal length and d the slit separation.

Fourier Optics Lab References

Charles S. Adams and Ifan G. Hughes. Optics f2f: From Fourier to Fresnel. 2018. doi:10.1093/oso/9780198786788.001.0001.